Work Trail — Copy/Paste Prompt for an AI Builder

Build a production-minded MVP named **Work Trail**. It is a field-workforce tracking and reporting product for Indian SMEs. Create a responsive **admin web dashboard** and an Android-first **employee mobile app**. Use a clean, trustworthy B2B visual style—not a surveillance-style UI.

Product rule

Location tracking is enabled **only after the employee taps Start work** and stops immediately when they tap End work. Make this clear in employee-facing UI. Never implement hidden or all-day tracking.

Preferred stack

- Monorepo: pnpm + Turborepo
- Admin: Next.js + TypeScript + Tailwind + shadcn/ui + TanStack Query + React Hook Form + Zod
- Mobile: React Native Expo development build + Expo Router + TypeScript + Zustand + TanStack Query + SQLite offline outbox
- API: NestJS + Prisma + PostgreSQL/PostGIS + Redis/BullMQ
- Auth: JWT access/refresh tokens with server-enforced roles
- PDFs: server-side HTML to PDF
- Use typed API contracts shared by web, mobile, and API

If the environment cannot create a mobile/native app, build the admin dashboard fully and create a documented React Native mobile-app folder with screens, services, hooks, types, and mocked/native adapters ready to run.

Roles

- ADMIN: create company, invite/manage employees, view employee map/updates/work hours, export reports.
- EMPLOYEE: log in, start/end only their own session, submit typed or voice-transcribed updates, view own history.

Admin routes

- `/` landing page
- `/login`
- `/signup`
- `/forgot-password`

- `/app/dashboard`
- `/app/employees`
- `/app/employees/new`
- `/app/employees/[id]` with Overview, Map, Updates, Work hours, Reports tabs
- `/app/reports`
- `/app/settings`

Employee screens

- Splash/session restore
- Login
- Permission disclosure
- Home/off-duty
- Active work session
- Add update
- Update confirmation/processing state
- History
- Profile/privacy/help

Required functionality

1. Admin signup/login, employee invitation with an expiring token, employee registration/login.
2. Admin dashboard: KPI cards for total employees, working now, checked-in today, total hours today, updates today; active employee list; recent updates.
3. Employee Start work creates exactly one active work session, starts elapsed timer, captures immediate initial location, and enables a persistent tracking indicator.
4. While active, target location collection every 5 minutes. Implement an adapter/service with local queue, retry, idempotency, and an Android background-task integration point. Clearly note OS restrictions and keep it best-effort.
5. Employee End work captures final location, ends session, calculates duration, and stops tracking.
6. Add Update page: multiline editable text field, microphone button which transcribes into the same editable field, category select, submit button. Store raw text and show AI-summary processing status.
7. Admin employee detail: date presets Today, Yesterday, This week, Last week, This month, Custom. Map tab shows chronological polyline, start/end markers, timestamps, accuracy warning, and data gaps. Updates tab shows raw updates and AI summary. Work hours tab shows sessions and totals.

8. Reports: generate/download a PDF with company/employee details, range, sessions, total hours, updates, and AI report. Protect downloads by role/company.
9. AI summary service: asynchronous queue, strict JSON-schema output, preserve raw updates, never invent facts, and show an AI-generated label.
10. Organisation isolation: every query is scoped to company ID and every API route checks role/ownership.

Data model

Implement: `companies`, `users`, `employees`, `invitations`, `work_sessions`, `location_points`, `work_updates`, `generated_summaries`, `reports`, and `audit_logs`.

Use UUIDs and UTC timestamps. Add a unique constraint preventing more than one active session per employee. Store device and server timestamps for location points. Use idempotency keys for batched location uploads.

Design direction

- Brand: **Work Trail**
- Personality: calm, accountable, field-friendly, modern B2B
- Primary: deep teal `#0F766E`
- Dark ink: `#0F172A`
- Surface: `#F8FAFC`
- Accent amber: `#F59E0B`
- Positive: `#16A34A`
- Warning: `#DC2626`
- Typeface: Inter or Manrope
- Use a left sidebar on desktop admin, bottom navigation for employee mobile.
- Use cards, clear status chips, generous spacing, accessible contrast, loading/empty/error states.
- Avoid aggressive red tracking imagery, dark hacker aesthetics, and misleading “live” language when data is not live.

Seed/demo data

Create one demo company, one admin, and 5 employees with realistic work sessions, location points around Mumbai, updates, work-hour totals, and generated summaries. Add a visible `Demo data` badge when using mocks.

Engineering requirements

- Environment variables in `.env.example`, no hard-coded secrets.
- OpenAPI/Swagger documentation.
- Zod validation at API boundaries.
- Unit tests for auth/authorisation, active-session rule, location ownership, and report access.
- Loading, empty, error, offline, and permission-denied UI states.
- README with local setup and physical Android location-testing steps.
- Generate these docs in `/docs`: `prd.md`, `plan.md`, `api-contract.md`, `data-model.md`, `ai-prompts.md`.

Do not build in MVP

Payroll, facial recognition, task assignments, geofencing alerts, route optimisation, employee scoring, hidden monitoring, or iOS-specific features.

Start by scaffolding the repo, database schema, auth, and a complete clickable admin dashboard using seeded data. Then connect real APIs in the order: invitation -> employee login -> start session -> location sync -> update -> admin review -> AI summary -> PDF.